Advanced Programming Practice · 21CSC203P

Unit 1 · Session 4 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 4 : SLO – 1 | Define Paradigms

Activity (Solutions)

• 1) Procedural: program = procedures operating on data — static int add(int a, int b){ return a+b; } OOP: program = objects with state+behavior — class Calculator { int add(int a,int b){ return a+b; } } Functional: computation = evaluation of pure functions — IntBinaryOperator add = (a, b) -> a + b;

• 2) Same logic both ways (area of rectangle) — Procedural: static double area(double l, double w){ return l*w; } called as area(4,5). Object-oriented: class Rect { double l,w; Rect(double l,double w){this.l=l;this.w=w;} double area(){ return l*w; } } called as new Rect(4,5).area().

• 3) Java features supporting multiple paradigms: classes/inheritance (OOP), static methods and control structures (procedural/imperative), lambda expressions, method references, functional interfaces and the Stream API (functional), and annotations/generics (declarative flavor).